



Certification 2017 Part 2 Version 1

ABT CERTIFICATION EXAMINATION

October 7, 2017

PART 2

This examination constitutes part A of the certification process for toxicologists attempting to become certified by the American Board of Toxicology. There are three parts to the exam. If you do not achieve a passing score on any of the parts, you will have to retake the examination. If you pass all three parts, your scores will be provided to you in November of this year, along with the results of the certification. If you have any questions, please contact the ABT office at (919) 841-5022 or info@abtox.org.

EXAMINATION CATEGORIES

Domain I. Design, Execute, and Interpret Toxicology Studies

Domain II. Descriptive Toxicology: Environmental, Clinical, Non-clinical, and Forensic Investigations

Domain III. Mechanistic Toxicology

Domain IV. Risk Assessment (Hazard Identification, Exposure Assessment, Dose Response Assessment, and Risk Characterization and Management)

Domain V. Applied Toxicology: Public, Environmental, and Occupational Health

INSTRUCTIONS

1. Use a soft (#2) pencil to fill the box corresponding to your answer.
2. Select only one answer for each question. (Erase others)
3. Do not leave any questions unanswered. (No penalty for guessing).
4. Sign answer sheet **ONLY** on the line that says name.
5. Do not bend, cut, staple or tape answer sheets.

Good Luck!

Certification 2017 Part 2 Version 1

© 2017 American Board of Toxicology, Inc. All rights reserved.
No part of this publication may be reproduced, stored in any retrieval system, or transmitted in any form or by any means, electric, mechanical, photocopying, recording or otherwise, without prior written permission of the American Board of Toxicology, Inc.

Certification 2017 Part 2 Version 1

*Key for the following questions
Choose the single most appropriate answer*

1. The acute toxicity study in animals provides ____.
A. an approximate lethal dose
B. information on target organs
C. information on dose selection for longer-term studies
D. all of the above
2. Intoxicept is a small molecule that has promise as an Alzheimer's therapeutic. In phase I clinical trials, the drug failed because plasma concentrations of the parent drug varied over 100-fold among the 30 volunteers. What is the MOST LIKELY cause of this unusually high variability of the concentration of parent drug in plasma?
A. high hydrophilicity
B. substrate for glutathione-S-transferase
C. substrate for cytochrome P450 2D6
D. molecular weight over 400 Daltons
E. extensive enterohepatic circulation
3. A manufacturer of cheese products would like to add a mold inhibitor to the paper used to wrap the cheese. He must first:
A. demonstrate the necessity of using a mold inhibitor
B. demonstrate that none of the inhibitor migrates into the cheese
C. provide data demonstrating the lowest effective level of inhibitor in the wrapper
D. provide animal toxicity data demonstrating that the inhibitor is non-toxic in animals at levels which are present in the wrapper
E. A and C
F. B and D
4. A plastic manufacturer that produces packaging for the pastry industry believes that the addition of compound z (its true name is a trade secret) to its plastic formulation will decrease oxygen permeability and increase shelf life. Before incorporating compound z into his plastics to be sold he must first:
A. demonstrate an increase in shelf life for pastry with compound z in the plastic wrap
B. demonstrate the levels of z found in the pastry items after being wrapped and stored
C. provide data demonstrating the lowest effective level of Z in the plastic
D. provide animal toxicity data demonstrating that compound Z is non-toxic at levels which are present in the wrapped food items
E. All of the above
5. Which of the following is (are) true in regard to the safety testing of non-GRAS food additives:
A. animal toxicity tests are conducted on all food additives
B. the level of use must not be in excess of the lowest effective level
C. tolerance limits are set based on a no-effect level obtained from animal toxicity data and includes a safety factor when possible of approximately 100-fold
D. tolerance limits for food additives which have been shown to be weak carcinogens in animal studies are set similarly as in (C) above, except that a safety factor of 1000-fold is used
E. All of the above
F. A, B, and C
G. A and C
H. B and D
6. Increasing the casein content of a partially purified diet up to a level of 36%
A. reduces the spontaneous hepatoma incidence below background in mice
B. increases the spontaneous hepatoma incidence above background in mice
C. has no effect on the incidence of mouse hepatomas
D. reduces the spontaneous incidence of kidney tumors below background in mice
E. reduces the spontaneous incidence of lung tumors below background in mice
7. Which of the following characteristics is of greatest importance in the development of a good pesticide?
A. a high LD₅₀ to the insect
B. a low LD₅₀ to mammals
C. prolonged bioavailability
D. high lipid solubility
E. a high degree of selective toxicity
8. Which of the following attributes is least important in the selection of a drug for clinical use?
A. margin of safety
B. relative potency
C. duration of action
D. route of elimination
E. efficacy

Certification 2017 Part 2 Version 1

9. In assessing reproductive function, the fertility index is calculated as the number of pregnancies (x 100) divided by the number of:

- A. resorptions
- B. viable pups born
- C. successful matings
- D. pups born
- E. estrus cycles

10. A dominant lethal study has been carried out using mice to determine the mutagenic effects of acute ip exposure to a compound. Which of the following effects could be indicative of a positive (mutagenic) effect from this study?

- A. reduced male fertility in weeks 1, 4, and 6 post-treatment
- B. reduced male fertility and a reduction in implantations in weeks 1, 2, and 3 post-treatment
- C. an increase in post-implantation death in weeks 1, 2, and 3 post-treatment
- D. both A and B
- E. both B and C

11. A compound is given to male mice on day 1. Using the serial mating technique, male fertility is severely depressed for the first 7 days but returns rapidly to normal and remains high for the rest of the 60-day test. The compound has most likely affected the:

- A. spermatogonia
- B. spermatocytes
- C. spermatids
- D. spermatozoa
- E. all of the above

12. Which statement is not true regarding the use of short-term mutagenicity assays for prediction of carcinogenesis?

- A. in many tests good correlations exist between in vivo activity and a positive in vitro response
- B. a positive response in one short-term in vitro assay should not be widely accepted as conclusive evidence for in vivo activity
- C. the relatively low cost and short time-to-results have been a major factor in their increasing popularity
- D. most mutagenicity assays have been shown to be accurate predictors of in vivo potency in mammalian species
- E. bacterial mutagenesis assays such as the Ames test have been more extensively validated than other tests

13. The Ames test:

- A. is a widely used screening test to detect mutagens
- B. is also known as the host-mediated assay
- C. utilizes salmonella strains which grow in the presence of histidine
- D. choices A and C are correct
- E. all of the above are correct

14. The laboratory species of choice for assessing skin contact allergens is the:

- A. rabbit
- B. rat
- C. dog
- D. mouse
- E. guinea pig

15. Which of the following parameters need not be monitored in order to interpret the results of a rat teratology study?

- A. fertility index
- B. number of corpora lutea in the female
- C. fetal body weight
- D. the occurrence of fetal resorptions
- E. the occurrence of dilated renal pelvis

16. The overall assessment of the teratogenic potential of chemical exposure on the developing rodent fetus is best made in terms of:

- A. malformations among the fetal population and embryoletality
- B. fetal toxicity and hazard assessment for the human population
- C. malformations among the fetal population and toxic effects in the pregnant dam
- D. embryo-lethality and toxic effects in the pregnant dam

17. Which testing regimen is best designed to assess teratogenic potential of chemical exposure on the developing rat fetus:

- A. exposure of the pregnant females during the period of major organogenesis assessment of three dosage levels, and inclusion of a concurrent, negative control group
- B. exposure of the pregnant females during the entire gestational period, exposure of the males prior to mating the females and inclusion of a concurrent, positive control group
- C. exposure of the pregnant females during the period of major organogenesis, assessment of three dosage levels, and inclusion of historical data from untreated animals
- D. exposure of the females prior to mating and through the period of major organogenesis, assessment of a single dosage level, and inclusion of a concurrent, negative control group

Certification 2017 Part 2 Version 1

18. A rat sacrificed after 20 months of a 2-year study was found to have gastric hemorrhage, enlarged rough kidneys, fine white streaks in the intestinal muscle and prominent white foci on the thyroids. The best tentative diagnosis is:

- A. secondary hypoparathyroidosis due to renal failure
- B. chronic progressive glomerulonephrosis
- C. gastric ulcers with septicemia
- D. nephroblastoma with disseminated metastasis
- E. none of the above

19. The data from a multigeneration reproduction study conducted in rodents are most effectively used to:

- A. assess the effects of chronic, low levels of chemical exposure on the reproductive capacity of the parental animals and on the growth and survival of the neonates
- B. assess the effects of high, toxic levels of chemical exposure on the reproductive capacity of the parental animals and on the behavioral development of the neonates
- C. assess the effects of acute exposure to chemicals on the survival of the parental and the neonatal animals
- D. assess the effects of chronic, low levels of chemical exposure on the reproductive capacity of the parental animals and the teratogenic effects in the neonates

20. Damage in which of the following tissues results in high Draize scores in the eye irritation test in rabbits?

- A. conjunctiva
- B. cornea
- C. iris
- D. lens
- E. retina

21. Genetic toxicology tests designed to detect clastogenesis include:

- A. Ames/Salmonella assay
- B. micronucleus test
- C. fungal aneuploidization
- D. chromosomal observation
- E. B and D

22. Which of the following *S. typhimurium* strains is sensitive to mutagens which induce pair substitutions?

- A. TA 1535
- B. TA 1537
- C. TA 1538
- D. all of the above
- E. none of the above

23. A rat with a respiratory rate of 100 breaths/min and a tidal volume of 1.3 mL has a minute volume equal to which of the following?

- A. 1.3 L/min
- B. 0.013 L/min
- C. 0.78 L/min
- D. 0.13 L/min
- E. none of the above

24. After inhalation exposure to an irritating dust, bronchoalveolar lung lavage revealed elevated alkaline phosphatase levels. This effect could be attributed to:

- A. overt hepatotoxicity
- B. release of lysosomal enzymes
- C. lung edema
- D. lung fibrosis
- E. damage of type I pneumocytes

25. An acute inhalation toxicity study is conducted by exposing 10 male and 10 female rats for one hour to the test agent. If the airflow is 10 L/min and the total weight disseminated into the chambers atmosphere is 120 grams, what is the nominal concentration (mg/L) for this exposure?

- A. 2
- B. 20
- C. 40
- D. 200
- E. 400

26. The theoretical time for a test compound to reach 99% equilibrium in an inhalation chamber for a selected target concentration is termed the T99. In addition to the equation constant, what measurements are needed to calculate the T99?

- A. chamber volume
- B. chamber target concentration of the test compound
- C. chamber airflow rate
- D. molecular weight of the test compound
- E. A and C
- F. B and D

27. The in vivo micronucleus test is often used in a standard battery of genetic toxicity assays. In this test, the micronuclei are defined as which of the following?

- A. pyknotic nuclei from a cell with decreased cytoplasmic to nuclear ratios
- B. clumps of ribosomes
- C. membrane-bounded structures that contain chromosomal fragments or whole chromosomes
- D. binucleated remnants of germ cells
- E. dense bodies of lysosomal origin

Certification 2017 Part 2 Version 1

28. Which of the following genetic toxicology assays measures both mutagenic and clastogenic endpoints?

- A. Ames test
- B. Mouse Micronucleus test
- C. Mouse Lymphoma assay
- D. Comet assay
- E. Sister Chromatid Exchange assay

29. Clinical assessment of drug-induced liver injury using Hy's Law (indication of hepatocellular damage in conjunction with jaundice in the absence of cholestasis) considers which of the following test results?

- A. >3-fold increases in ALT, AST, and ALP without increased total bilirubin
- B. >3-fold increases in ALT, AST, and ALP with >2-fold increases in total bilirubin
- C. >3-fold increases in ALT and AST (not ALP) and >2-fold increases in total bilirubin
- D. >3-fold increases in ALT, AST and >2-fold increases in total cholesterol
- E. >25-fold increases in ALP and >2-fold increases in total bilirubin

30. Which of the following is considered a reproductive physiology similarity among humans and rats?

- A. the menstrual cycle has a spontaneous luteal phase of 10 to 14 days
- B. FSH and LH regulated germ cell development after puberty
- C. pregnancy is easily disrupted by estrogens
- D. placental tissue expresses high levels of aromatase
- E. the spermatogenic cycle is approximately 75 days in duration

31. Which statement(s) is/are true for the Ames/Salmonella typhimurium assay?

- A. strains TA 1535 and TA 100 respond to mutagens that induce base-pair mutations
- B. tester strains TA 98 and TA 1537 contain the PKM 101 plasmid which increases the capacity of the bacteria to carry out mixed function oxidation reactions
- C. strains TA 1537, TA 1538 and TA 98 respond to mutagens that induce frame-shift mutations
- D. all strains have the normal DNA excision repair system
- E. A, B, and C
- F. A and C
- G. B and D

32. The initial disturbance of acid-base balance that occurs in individuals poisoned with very large doses of salicylates is the result of:

- A. metabolic acidosis
- B. respiratory alkalosis
- C. renal insufficiency
- D. metabolic alkalosis
- E. respiratory acidosis

33. All of the following may cause metabolic acidosis EXCEPT

- A. renal failure
- B. salicylates
- C. methanol
- D. diuretics
- E. diarrhea

34. All of the following are true of thallium except ____.

- A. Toxicity may result from mimicking sodium ion.
- B. Prussian blue is an antidote.
- C. It is one of the most toxic metals.
- D. Significant exposure can result in dermal, cardiac, or neurological toxicity.

35. Which of the following is the molecular basis for the acute effects of cyanide poisoning?

- A. formation of methemoglobin
- B. formation of carboxyhemoglobin
- C. inhibition of sulfhydryl enzymes
- D. inhibition of cytochrome oxidase
- E. inhibition of hydrolytic enzymes

36. All of the following are manifestations of lithium toxicity except ____.

- A. tremor
- B. liver necrosis
- C. diabetes insipidus
- D. renal failure

37. Sodium thiosulfate is useful in the antidotal treatment of cyanide poisoning because:

- A. it enhances the urinary excretion of CN-
- B. it inhibits "cyanide-thiosulfate transferase" and thereby blocks the formation of a toxic product from CN-
- C. it facilitates the formation of thiocyanate (SCN-) from CN-
- D. A and B
- E. none of the above

Certification 2017 Part 2 Version 1

38. The symptoms of Wernicke's encephalopathy may be caused by the lack of:

- A. folic acid
- B. glucose
- C. riboflavin
- D. thiamine
- E. niacin

39. Identification of anhydroecgonine methyl ester along with cocaine metabolites in the urine is an indicator that _____.

- A. Cocaine was adulterated with PCP.
- B. Cocaine was administered orally.
- C. Cocaine was smoked.
- D. Cocaine was taken with ethyl alcohol.

40. What is the purpose of including cytochalasin B in the in vitro micronucleus assay?

- A. to improve assay sensitivity by increasing cellular proliferation rates
- B. to block cytokinesis and avoid confusion associated with differences in cell proliferation kinetics
- C. to facilitate automated scoring by improving uptake of stains for micronuclei in the treated cells
- D. to decrease scoring errors by minimizing the incidence of binucleate and multinucleate cells
- E. to detect kinetochores and enable distinguishing between micronucleus formation by aneugenic versus clastogenic mechanisms

41. The use of ethanol or, preferably, fomepizole for alcohol dehydrogenase is a mainstay in the management of toxicity due to ingestion of methanol, ethylene glycol, or diethylene glycol. Which toxic metabolite of ethylene glycol may lead to anion-gap metabolic acidosis?

- A. formic acid
- B. oxalic acid
- C. formaldehyde
- D. glycolic acid

42. N-acetylcysteine (NAC) is a life-saving therapy in the management of poisonings with hepatotoxic compounds that may be detoxified by glutathione. The Rumack-Matthew nomogram is used to predict whether patients will develop hepatotoxicity based on the number of hours after ingestion (x-axis) and the plasma concentration (y-axis) for which of the following compounds:

- A. ibuprofen
- B. acetaminophen
- C. carbon tetrachloride
- D. Amanita phalloides

43. Movement of ionic organic compounds across gut or gills is strongly influenced by

- A. speciation
- B. lipid solubility
- C. pH partitioning
- D. diffusion capacity
- E. B and C are correct

44. Hydroxyl radical is enzymatically detoxified by _____.

- A. catalase
- B. glutathione peroxidase
- C. glutathione reductase
- D. none of the above

45. Which of the following are true for saturable plasma clearance described by Michaelis-Menten kinetics?

- A. first order when plasma concentration (C) is less than the Michaelis constant
- B. first order when C is much larger than the Michaelis constant
- C. zero order when C is less than the Michaelis constant
- D. zero order when C is equal to the Michaelis constant
- E. the Michaelis constant is not useful for describing concentration dependent kinetics

46. Peanut allergy is relatively common and can be quite severe in some individuals. Although food allergies that develop in childhood may be outgrown, peanut allergy is often lifelong. The primary mechanism of hypersensitivity to peanuts is which of the following?

- A. Type 1 hypersensitivity
- B. Type 2 hypersensitivity
- C. Type 3 hypersensitivity
- D. Type 4 hypersensitivity
- E. Type 5 hypersensitivity

47. All of the following are capable of accepting electrons from reductases and forming radicals except ____.

- A. paraquat
- B. doxorubicin
- C. n-hexane
- D. nitrofurantoin

Certification 2017 Part 2 Version 1

48. Which of the following statements is NOT CORRECT relative to toxicokinetic behavior of pharmaceuticals?

- A. xenobiotics eliminated in vivo via first-order processes are essentially completely eliminated
- B. a high apparent volume of distribution is indicative that the xenobiotic predominantly remains in plasma
- C. clearance of a xenobiotic from the body may encompass renal or biliary excretion, pulmonary exhalation, and/or hepatic biotransformation
- D. bioavailability describes the extent of absorption via extravascular administration to that determined following intravenous administration by a comparison of AUC values
- E. in a two-compartment model of a xenobiotic, a semi-logarithmic plot of plasma concentration vs. time demonstrates a biphasic slope

49. A common hematological manifestation of lead intoxication is a hypochromic microcytic anemia especially in children. The anemia is thought to result from a decreased life span of the erythrocytes and an inhibition of heme synthesis. What are the two major enzymes that are known to be inhibited by lead?

- A. d-aminolevulinate synthase, porphobilinogen deaminase
- B. d-aminolevulinate dehydrogenase, ferrochelatase
- C. porphobilinogen deaminase, uroporphyrinogen III cosynthase
- D. uroporphyrinogen decarboxylase, porphobilinogen deaminase
- E. uroporphyrinogen decarboxylase, ferrochelatase

50. A young suicidal patient has ingested an unknown substance, estimated at about 20 grams, about 2 hours ago and was rushed to the hospital. He is still conscious, and appears not to be in crisis. Calculations to determine both his anion gap (for diagnosing metabolic acids) and osmo-gap (for serum osmality) are normal. However, his liver transaminase values appear elevated, approaching the upper limit for normal. Excessively toxic levels of the ingested chemical were conclusively detected in his serum, after which the local poison control center has recommended immediate treatment with n-acetyl cysteine (NAC). What is the chemical he ingested and without proper and timely treatment and intervention, what is the eventual health consequence?

- A. aspirin: coma and renal failure
- B. methanol: irreversible neurological sequelae
- C. acetaminophen: hepatotoxicity
- D. toluene: neurologic and blood problems
- E. black tar heroin: addiction, plus neurological and behavioral decline

51. Resistance to the pharmacologic effects of some drugs can be rapidly acquired following only a few doses of the agent over a period of hours. Subsequent doses of the agent fail to elicit the response regardless of the size of the dose. This phenomenon is known as:

- A. tolerance
- B. idiosyncrasy
- C. physical dependence
- D. tachyphylaxis
- E. none of the above

52. The primary toxic target organ for toluene is:

- A. liver
- B. lung
- C. CNS
- D. adrenal gland

53. Photoallergic reactions of the skin to drugs:

- A. are characterized by an exaggerated sunburn of skin exposed to sunlight
- B. are elicited by much smaller doses of drugs than are phototoxic reactions
- C. are caused only by the topical application of drugs
- D. differ from phototoxic reactions in having an immunologic basis
- E. All of the above
- F. A and C
- G. B and D

54. Which of the following industrial chemicals are high-risk hazards for Type IV allergic reactions (cell mediated, delayed hypersensitivity)?

- A. trichlorethylene
- B. epoxy monomers
- C. vinyl acetate
- D. mercaptobenzothiazole
- E. All of the above
- F. A and C
- G. B and D

55. Minocycline, nicardipine, and spironolactone are examples of marketed drugs known to induce thyroid neoplasia in rats by increasing circulating levels of TSH through which of the following mechanisms?

- A. decreasing microsomal enzyme levels
- B. increasing the release of T4 from thyroglobulin
- C. microsomal induction resulting in decreased serum levels of T3, T4
- D. chemical inhibition of 5'-monodeiodinase
- E. increasing the organification of iodine through stimulation of thyroid peroxidase

Certification 2017 Part 2 Version 1

56. Which of the following agents specifically accumulates in hepatic stellate cells?

- A. iron
- B. vitamin A
- C. endotoxin
- D. acetaminophen
- E. pyrrolizidine alkaloids

57. Iron (Fe) is an essential mineral but overdoses result in damage to periportal hepatocytes. Which of the following best describes the reason for this specificity?

- A. excess Fe from the portal vein precipitates in the zone I hepatocytes causing mechanical disruption of the cells, and the concentration of Fe in the zone I
- B. Fe functions as an electron donor for the formation of reactive oxygen species leading to lipid peroxidation and the concentration of oxygen available to participate in this reaction is higher in the zone I region of the liver compared to the zone III region
- C. Fe is preferentially taken up by the zone I hepatocytes which overwhelms the ability of the cells to regenerate ATP via normal respiratory mechanisms
- D. excess Fe directly interferes in the formation of the cytoskeletal structure leading to inhibition of spindle formation and eventual loss of mitotic capability
- E. Fe is preferentially taken up in type I hepatocytes and inhibits ATP-dependent formation of bile leading to cholestasis in the zone I region

58. Chronic isopropyl alcohol exposure can potentiate carbon tetrachloride hepatorenal toxicity by:

- A. additive direct toxic effects
- B. induction of metabolism
- C. increased absorption
- D. decreased first-pass effect

59. The pesticide DDT exerts its toxic effects on the nervous system primarily by which of the following mechanisms?

- A. Blocking acetylcholinesterase at cholinergic synapses
- B. Inhibiting the release of GABA from nerve endings
- C. Blocking the efflux of calcium from nerve axons
- D. Blocking postsynaptic acetylcholine receptors
- E. Delaying the inactivation of sodium channels

60. _____ causes perturbations of thyroid hormone economy, most likely by an inhibition of sodium-iodide symporter. In chronic studies, this compound increased the incidence of thyroid follicular cell tumors when administered in drinking water to rats.

- A. sodium chlorate
- B. formate
- C. sodium dodecylsulfate
- D. thiouracil
- E. all of the above
- F. A, B, and C
- G. B, C, and D

61. A contemporary example of a chemical acting as a thyroperoxidase inhibitor is _____. This is a widely used antibacterial compound in food-producing animals with a current permissible tissue residue level of 100 ppb. Carcinogenicity studies at NCTR reported a significant increase of thyroid tumors in male Fischer 344 rats administered the high dose (2400 ppm) of this compound.

- A. rifampicin
- B. isoniazid
- C. chloramphenicol
- D. phenytoin
- E. sulfamethazine

62. Xenobiotics such as omeprazole or zidovudine can contribute to a deficiency of vitamin B12. Subsequently, this deficiency can result in changes in erythroid and myeloid lineages in bone marrow. These events can result in what type of anemia or condition?

- A. aplastic anemia
- B. sideroblastic anemia
- C. megaloblastic anemia
- D. iron-deficiency anemia
- E. pure red cell aplasia

63. Which of the following compounds has been suggested to produce liver injury via lipid peroxidation:

- A. dimethylnitrosamine
- B. ethylene dibromide
- C. phosphorus
- D. thioacetamide
- E. none of the above

Certification 2017 Part 2 Version 1

64. The primary area of the brain where MPP⁺ (the toxic pyridinium ion of MPTP, a contaminant of synthetic heroin) causes injury and death to dopaminergic neurons is:

- A. cerebral cortex
- B. choroid plexus
- C. substantia nigra
- D. tectum
- E. none of the above

65. A test article is an inducer of liver microsomal enzymes and produces thyroid adenomas after long-term administration to laboratory rats. Which of the following physiological alterations would be associated with these changes:

- A. thyroid follicular cell hyperplasia
- B. thyroid hypertrophy
- C. increased glucuronidation of thyroxine
- D. increased serum TSH
- E. all of the above

66. Once steady-state is attained during an inhalation, exposure to a vapor, uptake of the inhalant is primarily dependent upon:

- A. respiratory rate
- B. cardiac output
- C. tissue solubility
- D. metabolism
- E. none of the above

67. Which statement indicates that zero order pharmacokinetics are applicable for describing the elimination of a chemical from the body?

- A. the half-life increases with increasing dose
- B. the decline of the chemical from the body is not exponential
- C. the area under the plasma concentration vs time curve is proportional to dose
- D. the composition of excretory products may be changed both quantitatively and qualitatively with dose
- E. none of the above

68. A volatile chemical was shown to have a first order elimination half-life from the blood compartment of 35 minutes. Assuming zero order (constant rate) absorption, how long will it take to reach approximately 90% of the steady state blood level concentration upon exposure to this chemical via inhalation?

- A. 0.25 hours
- B. 0.50 hours
- C. 1.0 hours
- D. 2.0 hours
- E. 6.0 hours

69. If the elimination half-life of a compound is 2.31 hr⁻¹, what is the constant fractional rate of elimination? Assume first-order kinetics.

- A. 0.1
- B. 0.2
- C. 0.3
- D. 0.5

70. An easy way to determine the half-life of elimination is to plot the following variables in an appropriate manner. C_p is the plasma concentration, C_u is the urine concentration, and T is the time.

- A. plot $\log C_p$ v. T
- B. plot $\log C_u$ v. $\log T$
- C. plot C_u v. T
- D. plot C_u v. $\log T$
- E. A or B can be used

71. If an individual is administered 250 mg of a drug by iv, what % of the dose is eliminated after one hour. Assume the drug follows first-order kinetics and the elimination rate constant is 0.3 mg hr⁻¹

- A. 74%
- B. 26%
- C. 12%
- D. 5%

72. The clearance of a toxicant is 100 mL/min. The steady-state concentration in plasma is 0.4 mg/L. What is the intake rate?

- A. 40 mg/min
- B. 4.0 mg/min
- C. 0.4 mg/min
- D. 0.04 mg/min

73. An individual has a drug plasma concentration of 0.03 mg/L one hour after the drug was administered. The drug has a volume of distribution of 6 L. What is the total concentration of the drug in the individual's body?

- A. 0.18
- B. 0.36
- C. 0.72
- D. 1.44
- E. none of the above

Certification 2017 Part 2 Version 1

74. The median lethal dose (LD_{50}) is a measure of the acute toxicity of a drug or chemical. Which of the following is true regarding use of the LD_{50} ?

- A. the most reliable way to establish dose levels for a chronic toxicity study is to determine the LD_{50} dose of and then select a certain fraction of the LD_{50} as the as the top exposure level for the chronic test
- B. there is no consistent relationship between chronic toxicity and the LD_{50}
- C. the LD_{50} for laboratory animals is an accurate indicator Of the relative hazard of the chemical for man
- D. all of the above
- E. none of the above

75. The slope of a dose-response curve indicates:

- A. the mechanism of action of a chemical
- B. the sensitivity of response to dose
- C. the toxic hazard of a chemical
- D. the potency of a chemical
- E. the extent of absorption of a chemical

76. A probit plot of mortality data would provide a reliable estimate of the?

- A. no observed effect level in the test species
- B. dose that would kill 30% of the test species
- C. carcinogenic threshold dose for the test species
- D. dose that would kill 0.5% of the test species
- E. threshold limit value for the agent in man

77. For a chemical that can elicit a toxic response from an animal, the probability that the response will occur under stated exposure conditions can be formulated as a hazard index which is?

- A. constructed from an estimate of the total exposure that the animal will experience
- B. derived from a series of dosage vs. lethal response measurements leading to a precise LD_{50} for that species
- C. constructed by combination of indices of intrinsic toxicity and intensity of exposure for the chemical animal interaction
- D. derived from a toxicokinetic analysis of how the animal metabolizes, stores and excretes the chemical
- E. derived from a combination of the information in B and D

78. When dosage vs. response data for a chronic irreversible effect of a chemical in a target species are available at relatively high dosage levels:

- A. it is generally possible to estimate the percentage response at very low dose levels by linear extrapolation of the log dose vs response curve
- B. extrapolation to effects at low dosage levels is highly uncertain since the shape of the dose vs. response curve at these levels is generally unknown
- C. it is generally possible to extrapolate the dosage vs. response curve to determine by zero effect intercept a precise threshold dosage for the effect

79. If a NOEC of 100 mg/m^3 from a 90-day rodent inhalation study (6 hours of exposure per day, 5 days per week) is used for calculating a margin of exposure, what point of departure from the animal study would be used for assessing potential risks to workers over an 8-hour work day?

- A. 100
- B. 75
- C. 50
- D. 25

80. Drug A has an EC_{50} of 10 nM and a maximum effect of 95%. Drug B has an EC_{50} of 75 nM and a maximum effect of 100%. Drug C has an EC_{50} of 190 nM and a maximum effect of 34%. Which of the following is correct?

- A. Drug B is the most potent drug
- B. Drug A has the highest efficacy
- C. Drug C has the lowest efficacy but highest potency
- D. Drug A is the most potent drug
- E. Drug B shows the lowest potency

81. In the terms ED_{50} and EC_{50} , what is the main point of difference?

- A. ED_{50} is a median measure of effective dose of a drug in a population
- B. EC_{50} is a median measure of effective dose of a drug in an individual
- C. ED_{50} is measured to estimate the therapeutic index and EC_{50} is used to compare potency
- D. all of the above

Certification 2017 Part 2 Version 1

82. The therapeutic index for Drug A is $TD_{50}/ED_{50} = 60/20 = 3$, and for Drug B $TD_{50}/ED_{50} = 60/3 = 2$ and this would indicate:

- A. The drugs A and B kill before they are even effective
- B. Reasonable care with exposure is required to avoid toxicity with Drug B
- C. Reasonable care with exposure is required to avoid toxicity with Drug A
- D. The smaller the Therapeutic index the greater the relative safety

83. The applicability of a probabilistic risk assessment approach depends on the needs of decision makers and stakeholders. This is typically determined in a tiered fashion (*i.e.*, Tier 1 to Tier 3). Which of the following statements are correctly paired with the level of complexity based on the tiering system used?

- A. Tier 1 = moderate complexity, realistic exposure assumptions, more detailed analysis and/or modeling
- B. Tier 2 = screening level, conservative exposure assumptions, simple analysis, and or modeling
- C. Tier 3 = high complexity, probabilistic exposure assumptions, detailed site-specific analysis and/or modeling
- D. all of the above
- E. A and C
- F. B and C

84. Probabilistic risk assessment provides a point estimate, rather than estimates of the range and likelihood, of which of the following?

- A. hazard
- B. exposure
- C. risk
- D. mode of action
- E. A, B, and C
- F. none of the above

85. The planning and scoping phases of a risk assessment include identifying the purpose, scope, participants, approach, resources, and past experiences. Interrelated with these phases is problem formulation, which includes developing a conceptual model and an analysis plan. Which of the following considerations are those made for the conceptual model?

- A. sources
- B. stressors
- C. pathway/route
- D. receptors/endpoints
- E. data gaps
- F. all of the above
- G. A, B, C, and D

86. If the total daily intake of a specific chemical agent via drinking water is 0.1 mg/kg/day, and the NOAEL (no observed adverse effect level) for neurotoxicity is 100 mg/kg/day, which of the following is the MOE (margin of exposure) for neurotoxicity via the oral route?

- A. 10
- B. 0.001
- C. 10 mg/kg/day
- D. 1000
- E. MOE cannot be calculated without an uncertainty factor (UF)

87. The Acceptable Daily Intake (ADI) for pesticide residues:

- A. Is based on animal experiments and structure-activity relationships
- B. Is based on results of chronic dog studies applying a safety factor of 10
- C. Is applied only for fruits and vegetables
- D. Is based on animal and human health effects and no-observed-adverse-effect levels
- E. May not be exceeded in drinking water

88. For a rodent NOEC of 100 mg/m³ from a 90-day inhalation study in which the animals were exposed for 6 hours per day and 5 days per week, the adjustment to a human exposure scenario that encompassed continuous exposure (*i.e.*, 24 hours per day, 7 days per week) would equal ____ mg/m³:

- A. 18
- B. 35
- C. 285
- D. 560

89. If a risk assessor calculates a human equivalent dose of 100 mg/kg-bw/day, based on tumor incidence data from a cancer bioassay, what would the cancer slope factor be, assuming a benchmark response of 10%?

- A. $1 \text{ (mg/kg-bw/day)}^{-1}$
- B. $0.1 \text{ (mg/kg-bw/day)}^{-1}$
- C. $0.01 \text{ (mg/kg-bw/day)}^{-1}$
- D. $0.001 \text{ (mg/kg-bw/day)}^{-1}$

90. If you assume a screening level cancer risk of 10^{-6} , what would the slope factor be if the daily exposure is 10 mg/kg-bw/day and the cancer slope factor is $1 \times 10^{-3} \text{ (mg/kg-bw/day)}^{-1}$? Would this be an acceptable cancer risk based on the screening level benchmark of 10^{-6} ?

- A. 0.00001; yes
- B. 0.00001; no
- C. 0.01; yes
- D. 0.01; no

Certification 2017 Part 2 Version 1

91. In assessing the toxic hazard from a chemical dissolved in the water environment of an aquatic animal species, the probability of irreversible damage to the organism depends on:

- A. the animal absorbing and retaining some of the chemical
- B. the intrinsic toxic potency of the chemical or its metabolites in the animal
- C. the concentration of the chemical in the water
- D. the rate of biotransformation and elimination of the chemical in the animal
- E. All of the above
- F. A and C
- G. B and D

92. Which of the following toxic chemical effects can reduce the biomass of a population in a species for which the chemical is not directly toxic?

- A. killing of a symbiont
- B. killing of the species prey
- C. alteration of courtship behavior
- D. alteration of the surrounding plant community
- E. All of the above
- F. A and C
- G. B and D

93. Which of the following describe air quality standards (AQS) but not threshold limit values (TLV)?

- A. establish air pollution limits that are designed to protect man and his environment
- B. are based on an exposure of 40 hours per week
- C. are designed to protect the general population
- D. are established to protect the workers
- E. All of the above
- F. A and C
- G. B and D

94. Food web residue dynamics can best be studied by data from?

- A. structure-activity relationships
- B. chronic toxicity testing
- C. microcosm experiments
- D. daphnia toxicity tests
- E. stoichiometrics

95. Which of the following is the BEST approach to limiting solvent overexposure in an occupational setting?

- A. local exhaust ventilation
- B. personal protective equipment
- C. dilution ventilation
- D. process change/solvent elimination
- E. frequent job rotation

96. Which of the following is the biggest public health concern related to low level exposure to lead (e.g. blood level)?

- A. encephalopathy
- B. proximal tubular nephropathy in the elderly
- C. anemia
- D. cognitive delays and behavioral changes in children
- E. hemoglobin synthesis

97. Nanomaterials can enter target cells by passive diffusion, direct physical penetration, or active, receptor-mediated uptake by endocytosis or phagocytosis depending on their size and extent of agglomeration. Match the following NPs with how they enter target cells:

- A. small NPs up to 50 nm in diameter/passive diffusion
- B. SWCNTs/puncturing cells
- C. rigid, long, low aspect ratio NPs/frustrate phagocytosis
- D. all of the above
- E. none of the above
- F. A and C
- G. A and B

98. Which of the following methods of administration are acceptable for use quantitative risk assessment?

- A. intratracheal instillation (rats)
- B. oropharyngeal aspiration (mice)
- C. intranasal instillation (mice)
- D. inhalation (rats/mice)

99. Section _____ of the Clean Air Act currently lists 188 chemicals or classes of chemicals for which special standards and risk assessments are required. The chemicals listed are those of greatest concern on the basis of toxicity (including cancer) and estimated release volumes. Currently, there is a list of 33 Hazardous Air Pollutants (HAPs) from the list of 188 that are deemed to be of greatest concern, so-called a list of the "dirty thirty." Emissions of HAPs are mandated for control to the maximal achievable control technology (MACT), and any residual health risk after MACT is to be considered in a separate quantitative risk assessment.

- A. 112(b)
- B. 112(r)(1)
- C. 112(r)(7)
- D. 114

Certification 2017 Part 2 Version 1

100. The total body burden of a toxicant is 0.72 mg, and the toxicant's volume of distribution is 6 L. What is the plasma concentration of the toxicant that corresponds in time with the total body burden?

- A. 0.12
- B. 0.24
- C. 0.36
- D. 0.72
- E. none of the above